

MODULE:-1

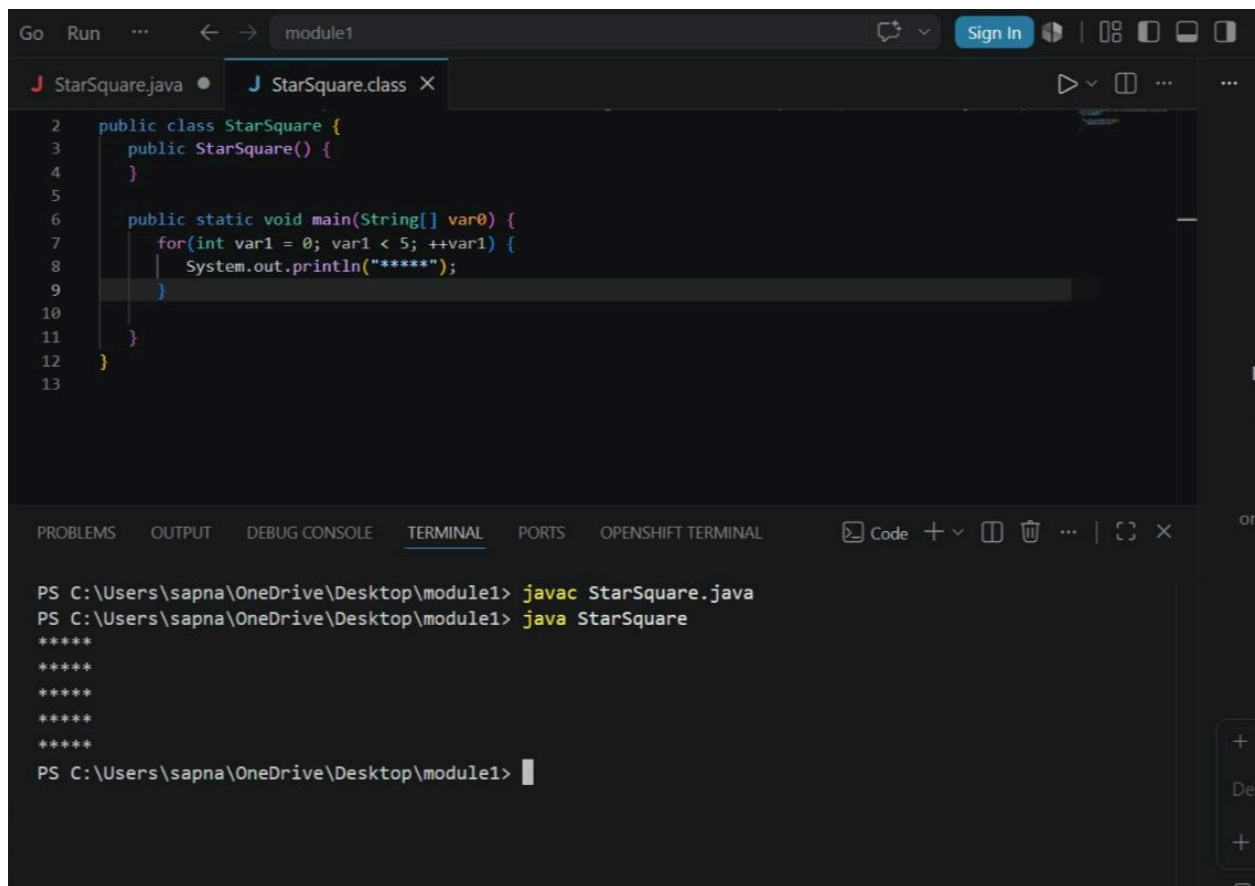
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Star pattern

1.Square



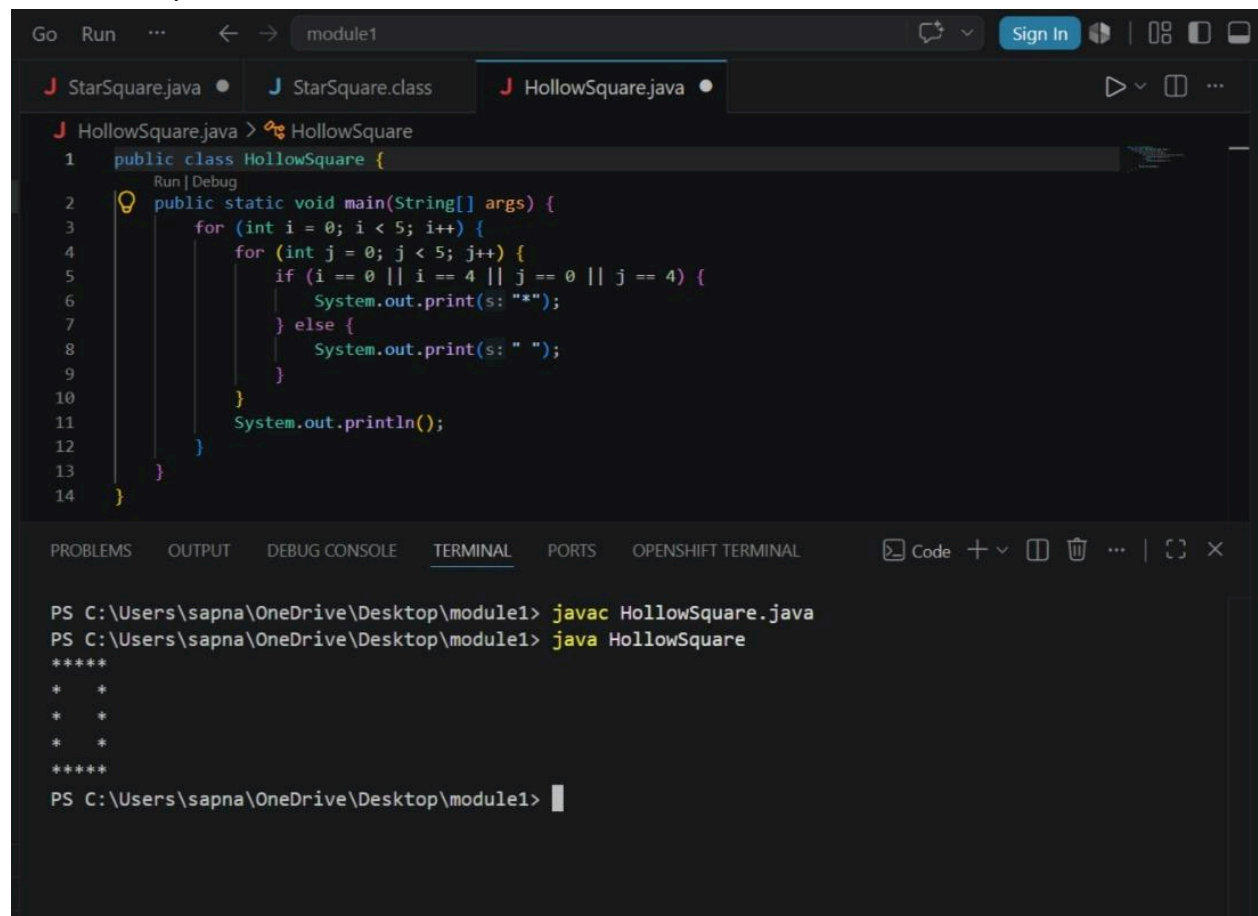
The screenshot shows an IDE with a Java file named `StarSquare.java` and its compiled class `StarSquare.class`. The code defines a `StarSquare` class with a `main` method that prints five lines of six asterisks each. The terminal at the bottom shows the compilation and execution commands, followed by the output of five lines of asterisks.

```
2 public class StarSquare {  
3     public StarSquare() {  
4     }  
5  
6     public static void main(String[] var0) {  
7         for(int var1 = 0; var1 < 5; ++var1) {  
8             System.out.println("*****");  
9         }  
10    }  
11 }  
12 }  
13
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS OPENSIFT TERMINAL

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac StarSquare.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java StarSquare  
*****  
*****  
*****  
*****  
*****  
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

2. Hollow Square



The screenshot shows an IDE with three tabs: StarSquare.java, StarSquare.class, and HollowSquare.java. The HollowSquare.java tab is active, displaying the following Java code:

```
1 public class HollowSquare {  
2     public static void main(String[] args) {  
3         for (int i = 0; i < 5; i++) {  
4             for (int j = 0; j < 5; j++) {  
5                 if (i == 0 || i == 4 || j == 0 || j == 4) {  
6                     System.out.print(s: "*");  
7                 } else {  
8                     System.out.print(s: " ");  
9                 }  
10            }  
11            System.out.println();  
12        }  
13    }  
14 }
```

Below the code editor, the terminal window shows the execution of the program:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac HollowSquare.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java HollowSquare  
*****  
*  *  
*  *  
*  *  
*****  
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

3. Triangle

The screenshot shows a Java IDE with a dark theme. The top toolbar includes buttons for 'Go', 'Run', and a 'Sign In' button. Below the toolbar, several tabs are open: 'numbers.java', 'HalfDiamondInverted.java 1', 'IncrementingDiamond.class', and 'StarTriangle.class'. The 'StarTriangle.class' tab is active, displaying the following code:

```
2 public class StarTriangle {
6     public static void main(String[] var0) {
7         for(int var1 = 1; var1 <= 5; ++var1) {
8             for(int var2 = 1; var2 <= var1; ++var2) {
10                }
11            }
12        System.out.println();
13    }
14 }
15 }
16 }
17 }
```

Below the code editor, the 'TERMINAL' tab is active, showing the execution of the program:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac StarTriangle.java
PS C:\Users\sapna\OneDrive\Desktop\module1> java StarTriangle
*
**
***
****
*****
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

4. Inverted Triangle

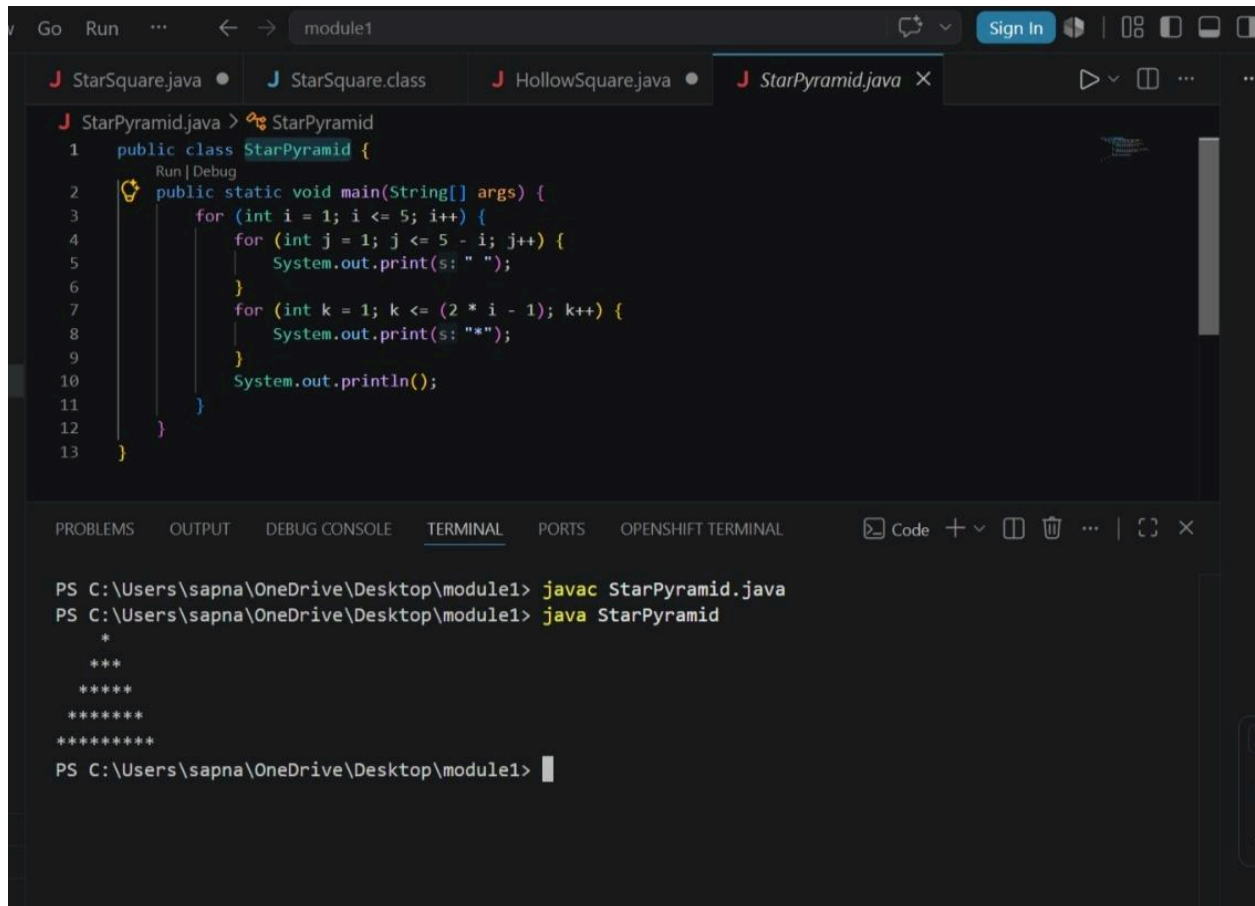
The screenshot shows an IDE with a dark theme. At the top, there's a toolbar with 'Go', 'Run', and other icons. Below it, a tab bar shows four files: 'StarSquare.java', 'StarSquare.class', 'HollowSquare.java', and 'InvertedTriangle.java'. The 'InvertedTriangle.java' file is active, displaying the following Java code:

```
1 public class InvertedTriangle {  
2     public static void main(String[] args) {  
3         for (int i = 5; i >= 1; i--) {  
4             for (int j = 0; j < 5 - i; j++) {  
5                 System.out.print(s: " ");  
6             }  
7             for (int j = 1; j <= i; j++) {  
8                 System.out.print(s: "**");  
9             }  
10            System.out.println();  
11        }  
12    }  
13 }
```

Below the code editor, there's a 'TERMINAL' tab. It shows the command prompt output for running the program:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac InvertedTriangle.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java InvertedTriangle  
*****  
****  
***  
**  
*  
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

5. Pyramid



The screenshot shows an IDE with a dark theme. The top toolbar includes 'Go', 'Run', and a 'Sign In' button. The file explorer at the top shows four files: 'StarSquare.java', 'StarSquare.class', 'HollowSquare.java', and 'StarPyramid.java'. The 'StarPyramid.java' file is open in the editor, displaying the following code:

```
1 public class StarPyramid {  
2     public static void main(String[] args) {  
3         for (int i = 1; i <= 5; i++) {  
4             for (int j = 1; j <= 5 - i; j++) {  
5                 System.out.print(s: " ");  
6             }  
7             for (int k = 1; k <= (2 * i - 1); k++) {  
8                 System.out.print(s: "**");  
9             }  
10            System.out.println();  
11        }  
12    }  
13 }
```

Below the editor is a terminal window with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', 'PORTS', and 'OPENSHIFT TERMINAL'. The 'TERMINAL' tab is active, showing the following commands and output:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac StarPyramid.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java StarPyramid  
*  
***  
*****  
*****  
*****  
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

6. Diamond

The screenshot shows an IDE with the following components:

- File Explorer:** Shows files `StarSquare.java`, `StarSquare.class`, `HollowSquare.java`, and `StarDiamond.java`.
- Editor:** Displays the code for `StarDiamond.java`. The code defines a `StarDiamond` class with a `main` method that prints a diamond pattern using nested loops. The pattern is symmetric around a center line.
- Terminal:** Shows the command `javac StarDiamond.java` and `java StarDiamond` being executed. The output displays the diamond pattern of stars.

```
1 public class StarDiamond {
2     public static void main(String[] args) {
3         int n = 5;
4         for (int i = 1; i <= n; i++) {
5             for (int j = 1; j <= n - i; j++) System.out.print(" ");
6             for (int k = 1; k <= (2 * i - 1); k++) System.out.print("*");
7             System.out.println();
8         }
9         for (int i = n - 1; i >= 1; i--) {
10            for (int j = 1; j <= n - i; j++) System.out.print(" ");
11            for (int k = 1; k <= (2 * i - 1); k++) System.out.print("*");
12            System.out.println();
13        }
14    }
15 }
```

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac StarDiamond.java
PS C:\Users\sapna\OneDrive\Desktop\module1> java StarDiamond
*
 ***
*****
*****
*****
 *****
 ***
 *
```

PS C:\Users\sapna\OneDrive\Desktop\module1>

7. Half Diamond

The screenshot shows an IDE with a Java file named `HalfDiamond.java`. The code defines a `HalfDiamond` class with a `main` method that prints a half diamond pattern using asterisks. The pattern consists of 5 rows: the first row has 1 asterisk, the second has 2, the third has 3, the fourth has 2, and the fifth has 1. The IDE's terminal window shows the command `javac HalfDiamond.java` followed by `java HalfDiamond`, which results in the printed pattern of asterisks.

```
1 public class HalfDiamond {
2     public static void main(String[] args) {
3         for (int i = 1; i <= 5; i++) {
4             for (int j = 1; j <= i; j++) System.out.print(s: "*");
5             System.out.println();
6         }
7         for (int i = 4; i >= 1; i--) {
8             for (int j = 1; j <= i; j++) System.out.print(s: "*");
9             System.out.println();
10        }
11    }
12 }
```

PS C:\Users\sapna\OneDrive\Desktop\module1> javac HalfDiamond.java
PS C:\Users\sapna\OneDrive\Desktop\module1> java HalfDiamond
*
**

**
*
PS C:\Users\sapna\OneDrive\Desktop\module1>

8. Half Diamond Inverted

The screenshot shows an IDE with a dark theme. The top toolbar includes 'Go', 'Run', and a 'Sign In' button. The file explorer shows four files: 'StarSquare.class', 'HollowSquare.java', 'StarDiamond.java', and 'HalfDiamondInverted.java'. The editor displays the code for 'HalfDiamondInverted.java' with line numbers 1 through 10. The code is as follows:

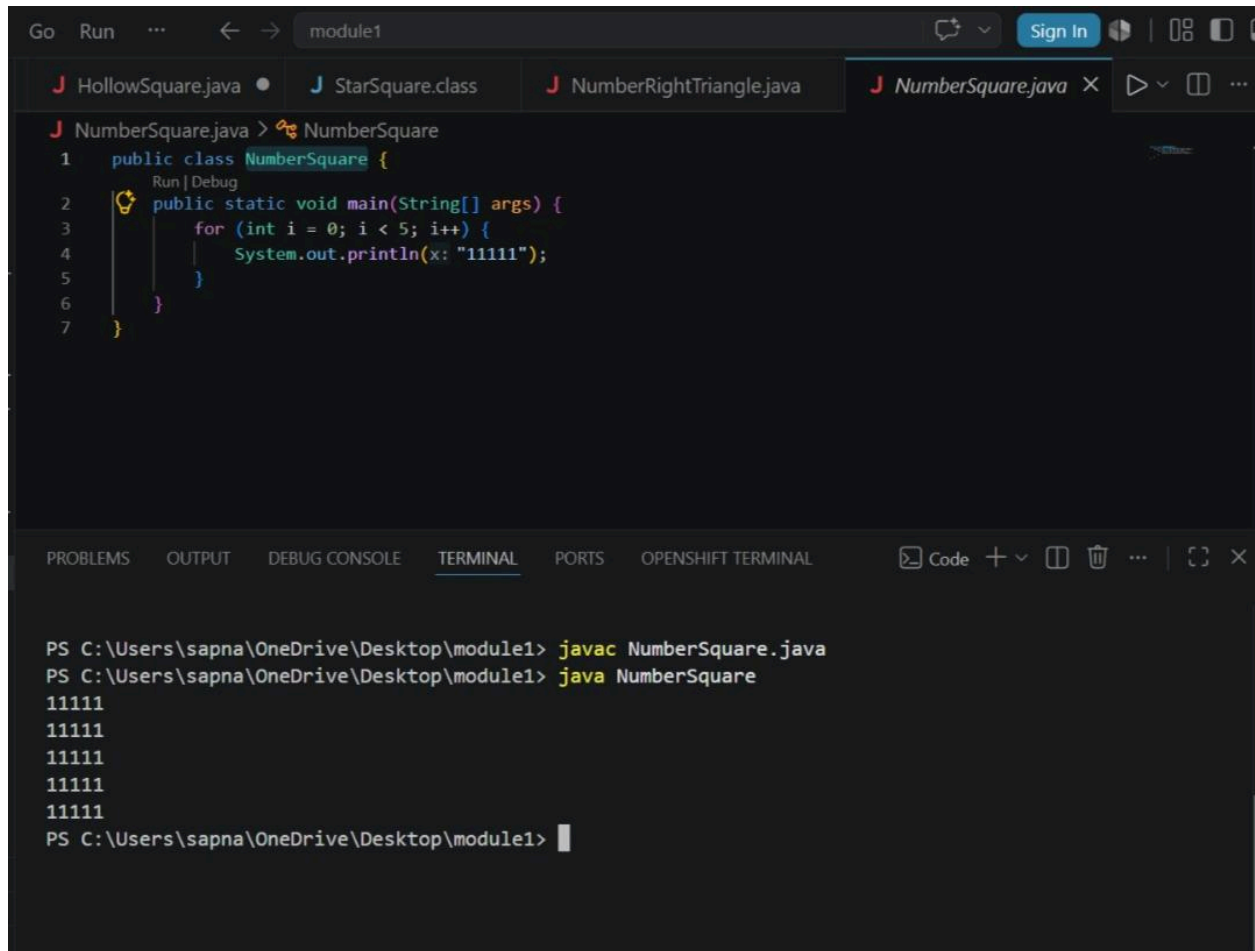
```
1 public class HalfDiamondInverted {  
2     public static void main(String[] args) {  
3         for (int i = 5; i >= 1; i--) {  
4             for (int j = 1; j <= i; j++) {  
5                 System.out.print(s: "");  
6             }  
7             System.out.println();  
8         }  
9     }  
10 }
```

The bottom panel shows the 'TERMINAL' tab with the following commands and output:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac HalfDiamondInverted.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java HalfDiamondInverted  
*****  
****  
***  
**  
*  
PS C:\Users\sapna\OneDrive\Desktop\module1> |
```

Number pattern

1. Square pattern



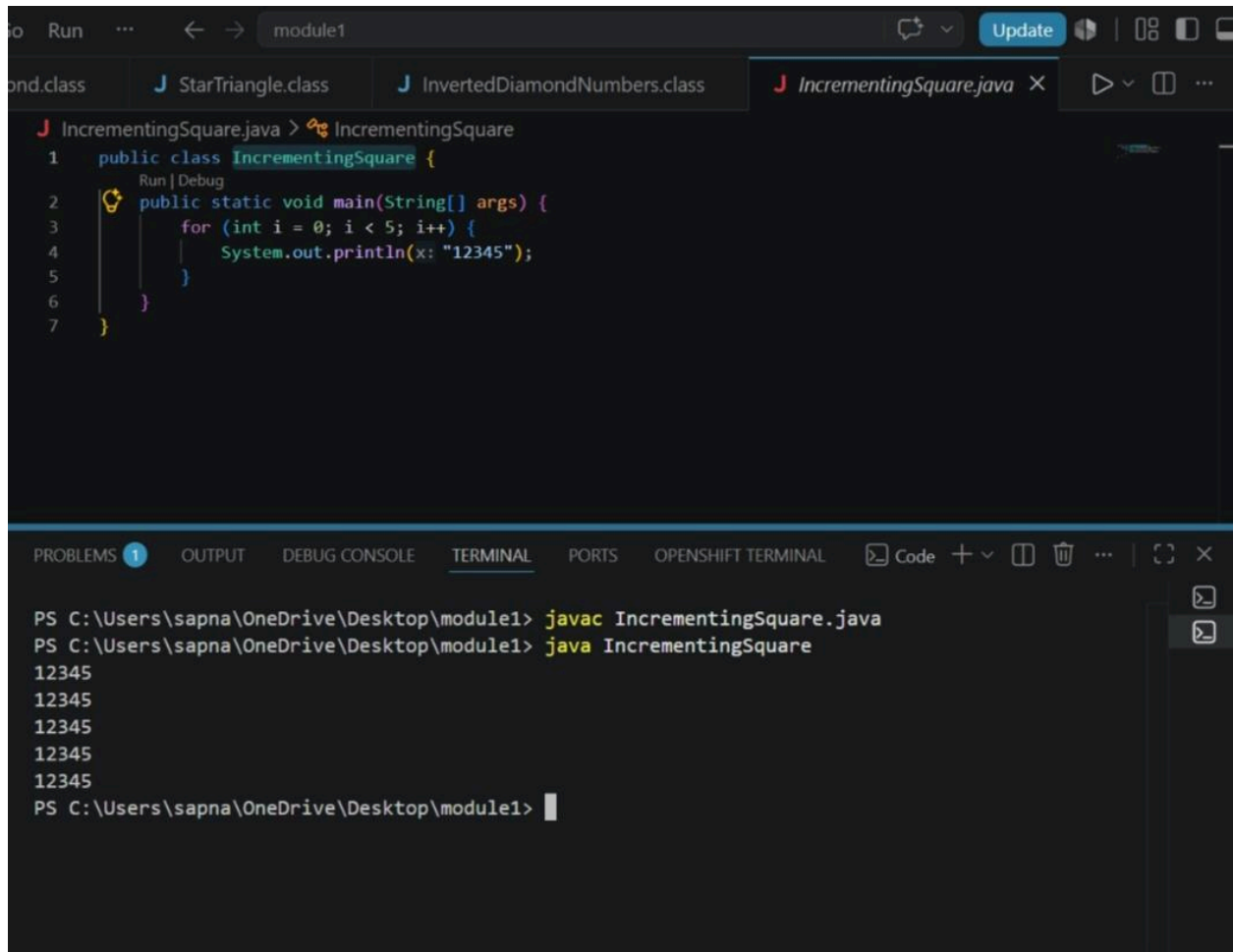
The screenshot shows an IDE window with a dark theme. The top toolbar includes 'Go', 'Run', and a 'Sign In' button. The file explorer at the top shows four files: 'HollowSquare.java', 'StarSquare.class', 'NumberRightTriangle.java', and 'NumberSquare.java'. The editor displays the code for 'NumberSquare.java':

```
1 public class NumberSquare {  
2     public static void main(String[] args) {  
3         for (int i = 0; i < 5; i++) {  
4             System.out.println(x: "11111");  
5         }  
6     }  
7 }
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac NumberSquare.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java NumberSquare  
11111  
11111  
11111  
11111  
11111  
PS C:\Users\sapna\OneDrive\Desktop\module1> |
```

2. Incrementing



The screenshot shows an IDE window with a tab for `IncrementingSquare.java`. The code defines a class `IncrementingSquare` with a `main` method that prints the string "12345" five times in a loop. Below the code editor, the `TERMINAL` tab is active, showing the compilation and execution of the program. The output of the program is five lines of "12345".

```
public class IncrementingSquare {  
    public static void main(String[] args) {  
        for (int i = 0; i < 5; i++) {  
            System.out.println("12345");  
        }  
    }  
}
```

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac IncrementingSquare.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java IncrementingSquare  
12345  
12345  
12345  
12345  
12345  
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

3. Right Triangle

The screenshot shows an IDE with a dark theme. At the top, there's a toolbar with 'Go', 'Run', and other icons. Below it, a tab bar shows four files: 'HollowSquare.java', 'StarSquare.class', 'NumberRightTriangle.java' (active), and 'NumberSquare.java'. The main editor displays the code for 'NumberRightTriangle.java':

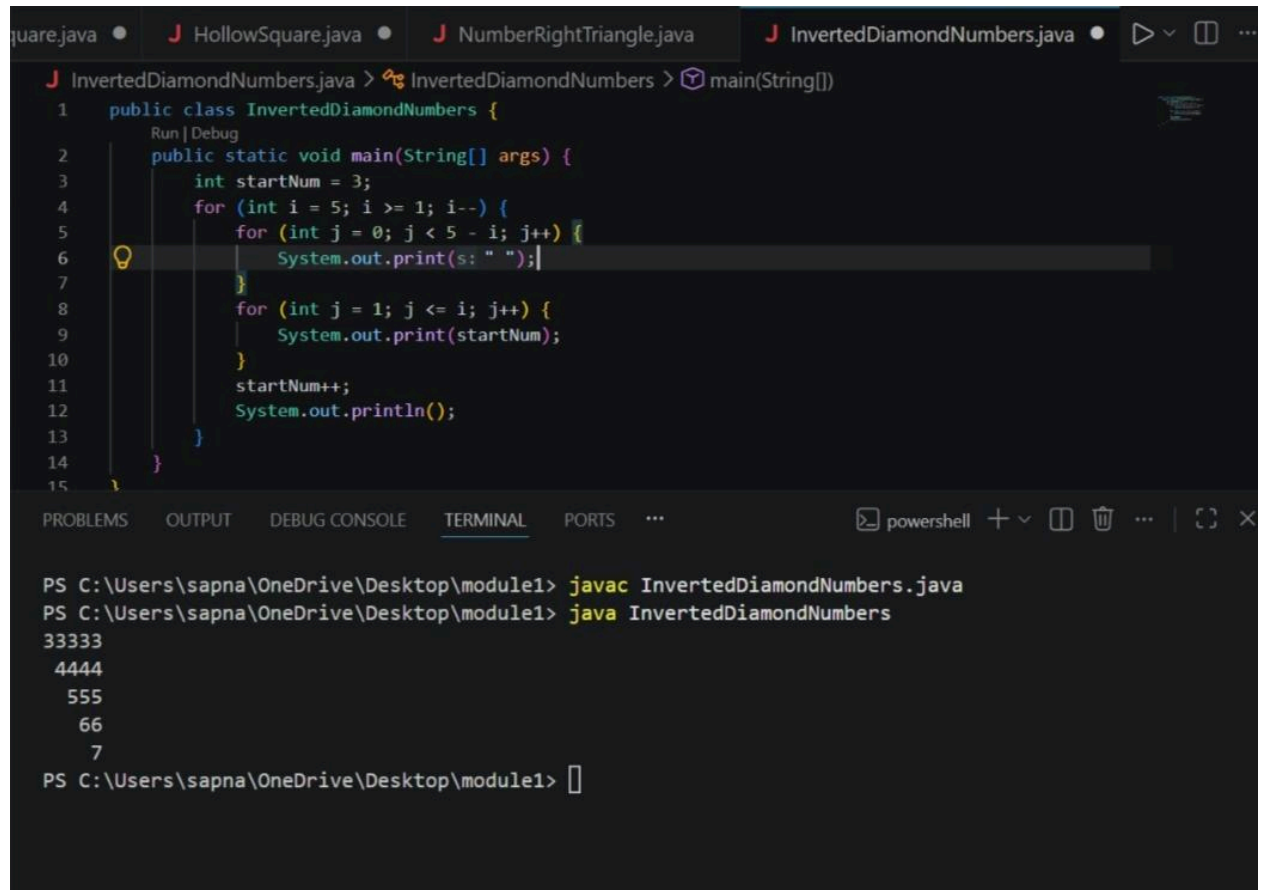
```
1 public class NumberRightTriangle {  
2     public static void main(String[] args) {  
3         for (int i = 1; i <= 5; i++) {  
4             for (int j = 1; j <= i; j++) {  
5                 System.out.print(j);  
6             }  
7             System.out.println();  
8         }  
9     }  
10 }
```

Below the editor is a terminal window with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (active), 'PORTS', and 'OPENSHELL TERMINAL'. The terminal shows the following commands and output:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac NumberRightTriangle.java  
PS C:\Users\sapna\OneDrive\Desktop\module1> java NumberRightTriangle  
1  
12  
123  
1234  
12345  
PS C:\Users\sapna\OneDrive\Desktop\module1> |
```

A 'Kill Terminal' button is visible in the top right corner of the terminal window.

4. Inverted Diamond



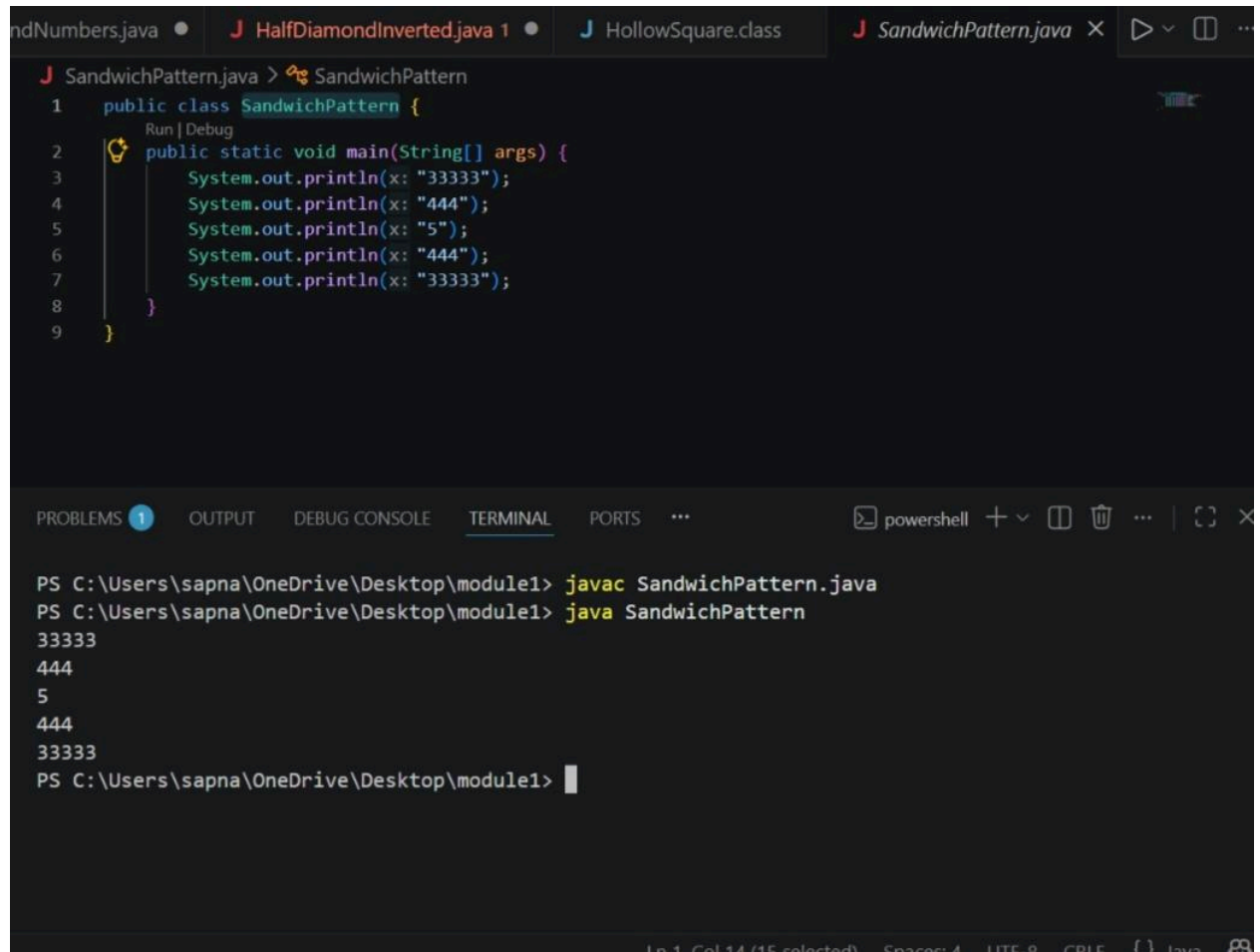
The screenshot shows an IDE with a Java file named `InvertedDiamondNumbers.java`. The code defines a `main` method that prints an inverted diamond pattern of numbers. The pattern consists of 5 rows, with the first row having 5 spaces followed by the number 3, and each subsequent row having one less space and one more number, until the last row which has 0 spaces and 5 numbers (3, 4, 5, 6, 7).

```
1 public class InvertedDiamondNumbers {
2     public static void main(String[] args) {
3         int startNum = 3;
4         for (int i = 5; i >= 1; i--) {
5             for (int j = 0; j < 5 - i; j++) {
6                 System.out.print(" ");
7             }
8             for (int j = 1; j <= i; j++) {
9                 System.out.print(startNum);
10            }
11            startNum++;
12            System.out.println();
13        }
14    }
15 }
```

The terminal output shows the execution of the program, resulting in the following pattern:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac InvertedDiamondNumbers.java
PS C:\Users\sapna\OneDrive\Desktop\module1> java InvertedDiamondNumbers
33333
 4444
   555
    66
     7
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

5. Sandwich pattern



The screenshot shows an IDE with the file `SandwichPattern.java` open. The code defines a class `SandwichPattern` with a `main` method that prints a sandwich pattern. The terminal output shows the pattern being printed.

```
1 public class SandwichPattern {
2     public static void main(String[] args) {
3         System.out.println(x: "33333");
4         System.out.println(x: "444");
5         System.out.println(x: "5");
6         System.out.println(x: "444");
7         System.out.println(x: "33333");
8     }
9 }
```

Terminal Output:

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac SandwichPattern.java
PS C:\Users\sapna\OneDrive\Desktop\module1> java SandwichPattern
33333
444
5
444
33333
PS C:\Users\sapna\OneDrive\Desktop\module1>
```

6. Incrementing Diamond Pattern

Go Run ... module1

numbers.java • J HalfDiamondInverted.java 1 • J HollowSquare.class J IncrementingDiamond.java X

J IncrementingDiamond.java > IncrementingDiamond

```
1 public class IncrementingDiamond {
2     public static void main(String[] args) {
3         for (int i = 3; i <= 7; i++) {
4             int count = i - 2;
5             for (int j = 0; j < count; j++) System.out.print(i);
6             System.out.println();
7         }
8         for (int i = 6; i >= 3; i--) {
9             int count = i - 2;
10            for (int j = 0; j < count; j++) System.out.print(i);
11            System.out.println();
12        }
13    }
14 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS ... powershell + -

```
PS C:\Users\sapna\OneDrive\Desktop\module1> javac IncrementingDiamond.java
PS C:\Users\sapna\OneDrive\Desktop\module1> java IncrementingDiamond
3
44
555
6666
77777
6666
555
44
3
PS C:\Users\sapna\OneDrive\Desktop\module1>
```